CHAPTER 10

Z-score & Robust Statistics

วัดว่า spread ห่างจากค่ากลางแค่ไหน — Standard vs Robust

KEY IDEA

Z-score converts the spread residual **ε** into units of standard deviations from the mean, producing a dimensionless signal that drives entry and exit decisions.

Standard $z = (\varepsilon - \theta) / \sigma_{stat}$

Robust $z = (\varepsilon - \text{median}) / (\text{MAD} \times 1.4826)$

The robust version protects against single outliers distorting the signal — critical when exchange glitches or thin-liquidity spikes contaminate the feed.

10.1 Standard Z-Score

Given a rolling window of **W** observations of the spread residual **ε**, the standard z-score is:

```
z_t = (ε_t − θ_W) / σ_W where: θ_W = (1/W) Σ ε_{t-i} for i = 0 … W-1 (rolling mean) σ_W = sqrt( (1/W) Σ (ε_{t-i} − θ_W)² ) (rolling std dev)
```

A common parameterisation uses **W = 24 hours** of data (e.g. 288 five-minute bars). Entry and exit thresholds map z-score values to actions:

Z-score range	Signal zone	Action
z > 2.0	Strong long signal	Enter long ε (buy cheap leg, sell expensive leg)
1.5 < z ≤ 2.0	Watch zone	Monitor; wait for confirmation
0.5 < z ≤ 1.5	Neutral-high	Hold existing position
−0.5 ≤ z ≤ 0.5	Neutral / exit	Close position (z < 0.5)
−1.5 < z < −0.5	Neutral-low	Hold existing short position
−2.0 < z ≤ −1.5	Watch zone	Monitor short signal
z ≤ −2.0	Strong short signal	Enter short ε (sell cheap leg, buy expensive leg)
|z| > 3.5	Stop-loss zone	Emergency exit — outlier or regime break

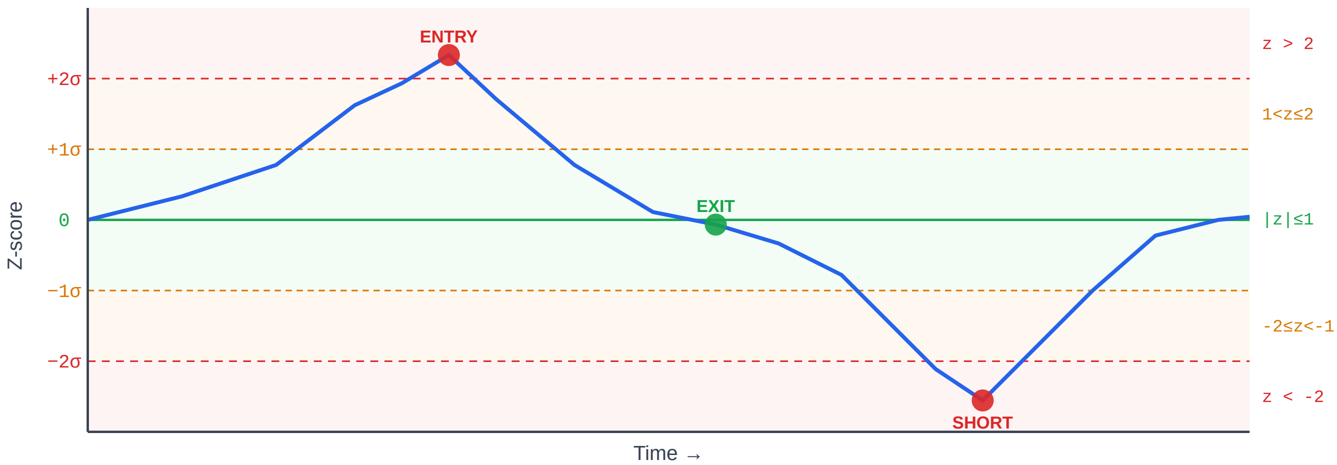


Figure 10.1 — Rolling z-score with colored signal bands. Blue curve = ε expressed in σ units. Red dot = entry signal (z > 2). Green dot = exit (|z| < 0.5).

10.2 The Outlier Problem

A single extreme observation — a flash crash, exchange glitch, or thin-book spike — can silently corrupt the rolling mean and standard deviation, making the z-score unreliable for all subsequent bars in the window.

Numerical Example

Suppose W = 10 bars with ε values (in %) before the outlier:

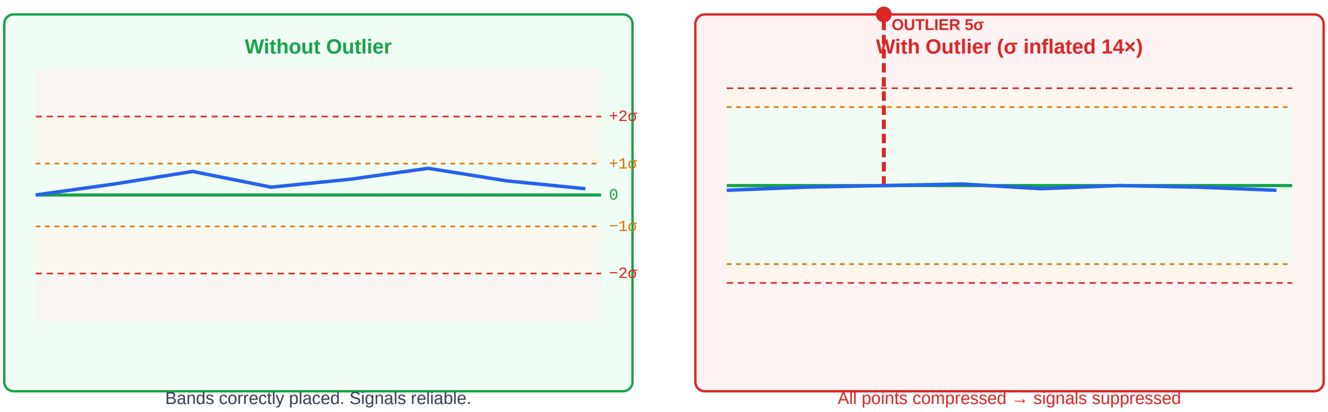
```
ε = [0.10, 0.12, 0.09, 0.11, 0.10, 0.13, 0.08, 0.11, 0.10, 0.12] θ_clean = 0.106%
σ_clean = 0.015% z for next point ε=0.22%: z = (0.22 - 0.106)/0.015 = 7.6 ← strong
signal
```

Now inject one outlier: replace bar 5 with ε = 0.80% (5σ spike):

```
ε = [0.10, 0.12, 0.09, 0.11, 0.80, 0.13, 0.08, 0.11, 0.10, 0.12] θ_dirty = 0.176%
σ_dirty = 0.212% ← mean shifted +0.07%, std blown up 14× z for same ε=0.22%: z =
(0.22 - 0.176)/0.212 = 0.21 ← noise, signal gone!
```

Warning: Outlier Contamination Effect

The outlier moved the mean by +0.07% (+0.47σ of the clean distribution) and inflated σ by roughly 14×. Every subsequent observation now has a z-score compressed toward zero — real signals are suppressed, and the bands become meaningless until the outlier leaves the window.



Bands correctly placed. Signals reliable.

All points compressed → signals suppressed

Figure 10.2 — Left: healthy z-score bands (σ ≈ 0.015%). Right: one 5σ outlier inflates σ and compresses all bands, hiding real signals.

10.3 Robust Z-Score via MAD

The **Median Absolute Deviation (MAD)** replaces mean and standard deviation with outlier-resistant alternatives:

```
Step 1: median_ε = median(ε₁, ε₂, …, ε_W) Step 2: MAD = median( |ε_t − median_ε| )
Step 3: σ_robust = MAD × 1.4826 Step 4: robust_z = (ε_t − median_ε) / σ_robust
```

Why 1.4826?

For a perfectly Normal distribution, MAD × 1.4826 = σ. The constant $1.4826 = 1 / \Phi^{-1}(0.75) \approx 1 / 0.6745$ makes MAD a consistent estimator of σ under normality, while remaining robust under contamination. A single outlier cannot materially change the median or MAD — it would need to corrupt more than 50% of the data.

Revisiting the same example with the 5σ outlier injected:

```
ε = [0.10, 0.12, 0.09, 0.11, 0.80, 0.13, 0.08, 0.11, 0.10, 0.12] median_ε = 0.11%
deviations from median = [0.01, 0.01, 0.02, 0.00, 0.69, 0.02, 0.03, 0.00, 0.01,
0.01] MAD = median(deviations) = 0.01% σ_robust = 0.01 × 1.4826 = 0.01483%
robust_z for ε=0.22%: (0.22 − 0.11) / 0.01483 = 7.42 ← signal preserved! (vs
standard z = 0.21 after outlier contamination)
```

Result

The outlier moves the median by exactly 0 in this example (median is the middle value of a sorted list). Robust z correctly identifies ε=0.22% as a strong signal (z=7.4), while standard z falsely showed no signal (z=0.21).

10.4 Rolling Window Z-Score

Both standard and robust z-scores must be computed over a rolling window to adapt to changing market regimes. The window length **W** is a key hyperparameter:

Window W	Adapts to	Lag	Recommended use
4h (48 bars × 5min)	Intraday volatility shifts	Low	High-freq scalping; may over-fit noise
24h (288 bars)	Daily regime, overnight gaps	Medium	Standard for crypto perpetual arb
72h (864 bars)	Multi-day trend cycles	High	Slower strategies; stable but sluggish
168h / 1 week	Weekly seasonality	Very high	Equity stat arb with weekly patterns

Lag Warning During Trend

If the spread drifts monotonically for several hours (e.g. during a BTC rally), the rolling mean θ_W chases the price upward with a lag equal to W/2. This artificially suppresses z-score readings — the strategy may fail to trigger exit on a position that is actually losing value. Consider adding a trend filter or using exponentially-weighted moving averages (EWMA) with $\lambda = 0.94$.

10.5 Comparison: Standard vs Robust Z-Score

Property	Standard Z	Robust Z (MAD)
Formula	$(\varepsilon - \text{mean}) / \text{std}$	$(\varepsilon - \text{median}) / (\text{MAD} \times 1.4826)$
Outlier sensitivity	High — one outlier shifts mean and std	Low — outlier must exceed 50% of data
Computational cost	O(W) — fast rolling sum/var	O(W log W) — requires sort for median
Efficiency (clean data)	100% efficient under Normality	~64% efficient under Normality
Best when	Data is clean, Gaussian, no spikes	Exchange feeds with occasional glitches
Use during	Stable, low-vol regimes	High-vol, crash periods, thin markets
Threshold calibration	Entry at $\lvert z\rvert > 2.0$	Entry at $\lvert z\rvert > 2.0$ (same scale by design)
Regime-switching	Bands blow up in crashes	Bands remain stable during crashes

Recommendation

Use **Robust Z as default** for crypto stat arb on Bybit/Lighter. Fall back to standard z only when your data pipeline has confirmed outlier filtering (e.g. price validated against two independent feeds). Hybrid: use robust z for entry, standard z for exit (since exit thresholds are softer and less sensitive to contamination).

Pseudo-Code

```
function compute_zscore(eps, theta, sigma): # Standard z-score (single-point, pre-computed θ and σ) if sigma == 0: return 0.0 return (eps - theta) / sigma function rolling_zscore(eps_series, W): # Compute standard rolling z-score for full series z_series = [] for t in range(W, len(eps_series)): window = eps_series[t-W : t] theta = mean(window) sigma = std(window) z_series.append(compute_zscore(eps_series[t], theta, sigma)) return z_series function robust_zscore(eps_window): # Compute robust z for the last point in eps_window med = median(eps_window) devs = [abs(x - med) for x in eps_window] mad = median(devs) sigma_r = mad * 1.4826 if sigma_r == 0: return 0.0 return (eps_window[-1] - med) / sigma_r function choose_signal(eps_window, W): # Use both; flag discrepancy std_z = rolling_zscore(eps_window, W)[-1] rob_z = robust_zscore(eps_window) discrepancy = abs(std_z - rob_z) if discrepancy > 1.5: log("OUTLIER SUSPECTED — using robust z") return rob_z, "robust" return std_z, "standard"
```

Running Example — Bybit BTC-PERP ↔ Lighter BTC-PERP (W = 24h)

Scenario: BTC crashes 12% over 2 hours at 03:00 UTC. Bybit order book thins out. One 5-minute bar records ε = 0.45% (a feed glitch as a single large market order clears thin asks).

Metric	Standard Z	Robust Z	Outcome
Window mean / median	θ = 0.126%	median = 0.10%	Robust unaffected by spike
σ estimate	σ = 0.060% (inflated 4×)	σ_r = 0.015%	Standard σ blown up
Z-score for next bar ε = 0.32%	z = (0.32−0.126)/0.060 = **3.2**	z = (0.32−0.10)/0.015 = **14.7**	—
Z-score for real signal ε = 0.22%	z = **1.6** (below entry threshold 2.0)	z = **8.0** (strong signal)	Standard misses entry
After outlier leaves window (24h later)	z recalibrates correctly	z unchanged (already correct)	Robust reacts immediately

Decision: During the BTC crash window, use **robust z for entry decisions**. The standard z produced a false alarm at ε=0.32% (z=3.2 suggesting strong signal, but driven by inflated σ from the spike) and missed the real signal at ε=0.22%. The robust z correctly ranked signals by true deviation from median spread.

Exercises

Exercise 10.1 — Compute Robust Z

Given the following window of 7 ε values (all in %):

```
ε = [0.10, 0.12, 0.08, 0.11, 0.80, 0.09, 0.11]
```

Compute: (a) median, (b) MAD, (c) σ_robust, (d) robust_z for the *last point* ε = 0.11%.

▶ Show Answer

Exercise 10.2 — Discrepancy Between Standard Z and Robust Z

You observe: **robust_z = 2.3**, **standard_z = 4.1**. Your entry threshold is |z| > 2.0.

(a) What does the discrepancy of 1.8 σ-units suggest? (b) Should you enter the trade? (c) What additional check would you perform?

▶ Show Answer